UM-2 solution

An accurate approximate solution of the coupled field equations, with Kohn-Sham theory for the dirac16complex field coupled to gravity and to the Standard-Model fields

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1 How to read this document

This document presents the approximate solution of the coupled field equations of UM-2 and the verification of each step. It is self-contained: every equation it uses is written here. Labels: AUTHOR, RECORD, STANDARD,

ASSUMED, INVENTED, DERIVED (computed and verified; the verification is stated), ESTIMATE, OPEN. Units $\hbar = c = 1$; in the numerical work $H = 1$ and $m = H$ unless a physical unit is given. The coordinates are the author's $x_0, x_1, \ldots, x_7$ (x_0 the hidden space direction, x_1, x_2, x_3 space, x_4 the time, x_5, x_6, x_7 the extra times); W^+ carries the electric charge $+e$ and W^- the charge $-e$.

Symbols with more than one use: k is the block variable $1/f^2$ (k_*, δk, $G(k)$), the Lovelock order as the index of α_k, $E_{(k)}$, $\mathcal{L}_{(k)}$ (written p where the block variable appears in the same formula), the momentum of a mode along x_1, x_2, x_3 in h_j, $\vec{k}$, Δk and $\varepsilon = e^{-a_4}|k|$, and the comoving Fourier wavenumber in the perturbation equations of step 5. $\Lambda = -10H^2$ is the eight-dimensional cosmological constant and Λ_4 the four-dimensional one; in the x_0 eigenvalue problems of step 2, $\Lambda = (m\,fL)^2$ is a dimensionless eigenvalue; $\Lambda_\star$ is the scale of the effective interactions. n is a number density, except as a level number ($n\pi$) and in the lattice vector $\vec{n} \in \mathbb{Z}^3$ of a momentum shell; N is a particle number. The subscript t (p_t, v_t) refers to the extra times x_5, x_6, x_7; $\xi = y/L$ is the scaled hidden coordinate; primes denote $d/d\xi$ in step 2 and $d/d\tau$ (conformal time) in step 5. f is the block scale and f_o an occupation number. ρ is an energy density, except for the electroweak parameter $\rho = \Lambda_W/(\Lambda_Z \cos^2\theta_W)$ (measured value ρ_0), and θ_W is the angle of the gauge couplings. w denotes the Lovelock weights (w_2, w_3), the weight $1/2$ of a fold parity in the Kohn-Sham sums, $\sqrt{c^2 - e^2}$ in the fermion mass condition and an equation-of-state parameter. ψ is a Standard-Model fermion field and, in step 5, the Newtonian potential; σ is a Pauli matrix ($\sigma_1, \sigma_2, \sigma_3$), the annihilation cross-section (σv), σ_{SI} the scattering cross-section, σ_b the cosmic variance of a bin of multipoles and, in step 5, the shear; t_o is an orbital density.

2 The equations being solved

The action of UM-2 is

$$S = \frac{1}{2\kappa}\int d^8x\sqrt{|g|}\Big(\sum_{k=1}^{3}\alpha_k\mathcal{L}_{(k)} - 2\Lambda\Big) - \int d^8x\sqrt{|g|}\,\rho_b(f) + S_\Psi + S_{gauge} + S_f + S_{tip} + S_{I9} + S_{I11} + S_{I12},$$

with the author's $\alpha_k = w_k H^{2-2k}$, $w = (1, \frac{1}{32}, \frac{1}{1152})$, $\Lambda = -10H^2$; the stabiliser $\kappa\rho_b = \kappa\rho_b^*(f_*/f)^8$; the dirac16complex Lagrangian $\sqrt{|g|}[\frac{1}{2}(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi) - m\bar\Psi\Psi - \frac{\lambda}{2}(\bar\Psi\Psi)^2]$ with the canonical spin connection; the eight-dimensional $SU(3)_c \times SU(2)_L \times U(1)_Y$ gauge fields and one generation of fermions; the tip sigma field $\Sigma \in SU(2)$ (no Higgs particle); and the portals

$$S_{I9} = \int_{y=-L} d^7x\sqrt{|g_{tip}|}\,(c_Z + c'_Z\varsigma)(i\bar\Psi\gamma^\mu P_+\Psi)\mathcal{Z}_\mu, \qquad S_{I11} = \int d^8x\sqrt{|g|}\,\frac{c_V}{\Lambda_\star^6}(i\bar\Psi\gamma^M\Psi)\sum_\psi q_\psi(i\bar\psi\gamma_M\psi),$$

$$S_{I12} = -\int d^8x\sqrt{|g|}\,\frac{\bar\Psi\Psi}{\Lambda_\star^3}\big(c_G G^\alpha_{MN}G^{\alpha MN} + c_W W^a_{MN}W^{aMN} + c_B B_{MN}B^{MN}\big),$$

$\mathcal{Z}_\mu = \frac{iv_\Sigma^2}{2}\mathrm{tr}(\Sigma^\dagger D_\mu\Sigma\tau^3) \to \frac{v_\Sigma^2}{2}\sqrt{\hat g^2 + \hat g'^2}Z_\mu$, $P_+ = (1+\Gamma_4)/2$. The field equations are

$$\sum_k \alpha_k E_{(k)}{}^M{}_N + \Lambda\delta^M_N = \kappa T^M{}_N, \qquad \gamma^M D_M\Psi = \Big[m + \lambda\bar\Psi\Psi + \frac{\mathcal{F}}{\Lambda_\star^3}\Big]\Psi - i\frac{c_V}{\Lambda_\star^6}\mathcal{J}_M\gamma^M\Psi - i\delta_{tip}(c_Z + c'_Z\varsigma)\mathcal{Z}_\mu\gamma^\mu P_+\Psi,$$

together with the Yang-Mills equations with fermion, tip and portal currents, the Dirac equations of the fermions with the tip Yukawa terms, and the boundary conditions at the fold and the tip.

3 The approximation scheme

The solution uses the small parameters of the problem:

- $\epsilon \sim 10^{-5}$: the metric is $g = \bar g + \epsilon h^{(1)} + \ldots$ (order 0: the background; order 1: the CMB, CNB, BAO);
- $(A/H)^2 \ll 1$ after the primordial era: the matter sources are tiny compared with the stabiliser (today 10^{-92}), so the block responds adiabatically and linearly;
- the Standard-Model masses are far below the gap $3H/(2f)$ of the x_0 operators: four-dimensional effective fields;

- dirac16complex is treated by Kohn-Sham density-functional theory on the static block, with the portal couplings as external mean fields (the Standard-Model fields enter through their densities and condensates).

The hierarchy: (1) the background and the primordial era; (2) the x_0 spectra, i.e. all masses; (3) the Kohn-Sham states of Ψ coupled to gravity and to the Standard-Model fields, and their back-reaction on the block; (4) the electroweak sector, N_{eff} and the CNB; (5) the linear cosmology; (6) the dark-matter relic.

4 Step 1: the background

4.1 The static block

The order-0 metric is

$$ds^2 = e^{2a_4}d\vec{x}^{\,2} - dx_4^2 + f^2\big[dy^2 - e^{2Hy}d\vec{x}_t^{\,2}\big], \qquad y = \frac{\ln|\sin 6Hx_0|}{6H}, \quad f = e^{-a_4}b_4^{1/2},$$

periodic in x_0 with period $\pi/(6H)$ (two halves joined at the fold $y=0$, a regular tip at $y=-L$, $\eta = HL = 3$). With $k = 1/f^2$, $A = \dot{a}_4$, $h = \dot{f}/f$, a static block ($h=0$, $A=0$) with the stabiliser ($p_3 = -\rho_b$, $p_0 = p_t = \rho_b$) requires (DERIVED)

$$\kappa\rho_b = 10H^2 - 6H^2k + \tfrac{3}{8}H^2k^2 = 3H^2k - 10H^2 \implies k_* = 12 + \tfrac{4}{3}\sqrt{51} = 21.5219,$$

$$f_* = 0.215556, \qquad G(k_*) = 1 - 24w_2k_* + 72w_3k_*^2 = \tfrac{20}{3} + \sqrt{51} = 13.8081 > 0,$$

$$\kappa\rho_b^* = (26 + 4\sqrt{51})H^2 = 54.566H^2.$$

$G(k_*) > 0$: four-dimensional gravity is attractive with the author's weights and Λ unchanged. (Without the stabiliser the only static block with $A=0$ is $k_0 = 8$ with $G = -1$, repulsive.) The linearised system has the eigenvalues $0, \pm i\omega_r$,

$$\omega_r^2 = \Big(\frac{31552}{2249} - \frac{2416\sqrt{51}}{6747}\Big)H^2, \qquad \omega_r = 3.38705H,$$

so the block is stable; the radion mass is $m_r = 3.387H = 48.42$ TeV. The whole stabiliser family $p_0 = \nu\rho_b$ ($\nu = 1/4$ to 3) gives stable blocks with $\omega_r = 3.33H$ to $4.10H$ (DERIVED).

4.2 The primordial era: the deflating extra times

The full nonlinear equations (all three components, the constraint preserved to 6×10^{-10}) were integrated from an expanding 3-space and a block larger than the stabilised one ($k_0 < k_*$) that deflates at $h_0 = -H/2, -H/3, -H/10$, for k_0 from 1.5 to 20 (DERIVED):

start k_0	$h_0 = -H/10$	$h_0 = -H/3$	$h_0 = -H/2$
1.5	singular at $x_4 = 1.44/H$	singular at $1.26/H$	singular at $1.14/H$
2.3	singular at $0.33/H$	singular at $0.31/H$	no expanding start
2.7	$k(60/H) = 21.415$	21.435	21.451
3	21.412	21.434	21.455
6	21.405	21.453	21.500
10	21.446	21.541	21.584
20	21.634	21.640	21.638

Every start above the separatrix $k_c = 2.5301, 2.5991, 2.6983$ (for $h_0 = -H/10, -H/3, -H/2$; just above the unstable static block $k_- = 2.4781$) ends at the stable block: the extra times x_5, x_6, x_7 and the hidden direction x_0 deflate at rates up to $|h| = 2.63H$ (in the stored runs $k_0 = 3, 5, 10$), by up to $f(k_c)/f_* = 2.917$, reach k_* within $x_4 < 1/H$, and then oscillate about it with the radion frequency ($3.3868H$ measured, $3.3870H$ linear), while 3-space keeps expanding; the oscillation energy redshifts like dust ($A\,x_4 \to 0.667$ to 0.670; 2/3 for dust). The extra times therefore deflate exponentially only in a short primordial era and are then frozen: the masses, all of which depend on f, become constant.

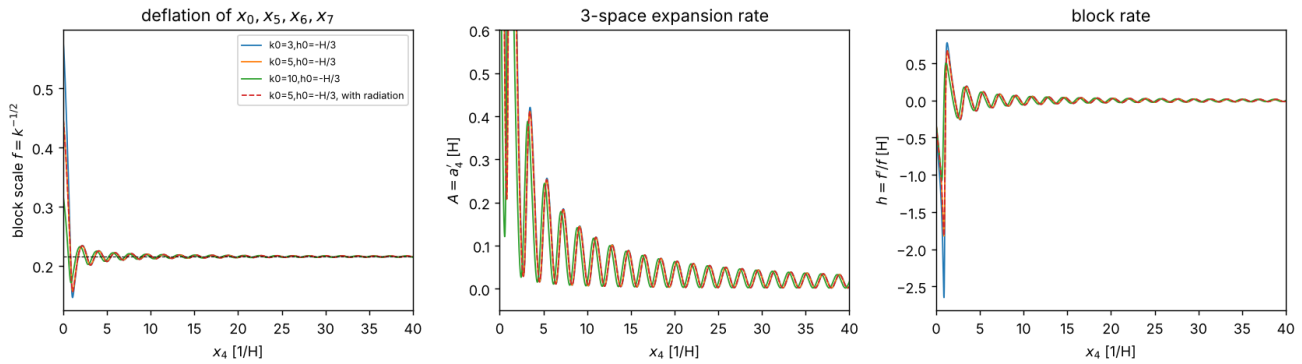


Figure 1: The primordial era: the block scale f, the 3-space expansion rate A and the block rate h for starts $k_0 = 3, 5, 10$ at $h_0 = -H/3$ (one with radiation), from the full nonlinear Einstein-Lovelock equations with the author's weights.

4.3 The response of the block to matter, and four-dimensional gravity

To first order about the static block with block-averaged matter sources ($\bar{p} = (p_0 + 3p_t)/4$; A^2 of first order, $3G(k_*)A^2 = \kappa\delta\rho$) (DERIVED):

$$\delta\ddot{k} + \omega_r^2\,\delta k = -2k_*\big(s_{h3}\,\kappa\delta p_3 + s_{hp}\,\kappa\delta\bar{p} + q_h\,\kappa\delta\rho\big), \qquad s_{h3} = 0.028666, \quad s_{hp} = 0.037320, \quad q_h = -0.009555;$$

the adiabatic shift is $\delta k = (-0.107556\,\kappa\delta p_3 - 0.140028\,\kappa\delta\bar{p} + 0.035852\,\kappa\delta\rho)/H^2$, and 3-space obeys $\delta\dot{A} = -\kappa(\delta\rho + \delta p_3)/(2G(k_*))$. Together with the constraint $3G(k_*)A^2 = \kappa\rho$ and the dimensional reduction ($\sum_p \alpha_p \mathcal{L}_{(p)} - 2\Lambda - 2\kappa\rho_b = G(k)R^{(4)} + \text{topological} - 2G(k)\Lambda_4$, p the Lovelock order), the four-dimensional gravity of UM-2 is general relativity,

$$G^{(4)}_{\mu\nu} + \Lambda_4 g_{\mu\nu} = 8\pi G_N T^{(4)}_{\mu\nu}, \qquad 8\pi G_N = \frac{\kappa}{G(k_*)V_\perp}, \qquad V_\perp = \frac{2f_*^4 v_t(1 - e^{-3\eta})}{3H},$$

with $\Lambda_4 = [6H^2k(1 - 2w_2k) + \Lambda + \kappa\rho_b]/G(k) = 0$ at the static point; the observed dark energy needs $\Lambda_4/H^2 \sim (H_0/H)^2 = 1.0 \times 10^{-92}$ (tuned; OPEN).

5 Step 2: the masses from the x_0 dependence of the wave functions

Every four-dimensional mass is an eigenvalue of the x_0 operator of its field on the static block; the eigenfunctions depend on x_0 through $y(6Hx_0)$ and are periodic with period $\pi/(6H)$ (the author's requirement). With $\xi = y/L$:

5.1 Gauge bosons and the graviton

$$-e^{-3\eta\xi}\big(e^{3\eta\xi}u'\big)' = \Lambda u, \qquad \Lambda = (m\,fL)^2, \qquad u'(0) = 0 \text{ (even) or } u(0) = 0 \text{ (odd)}, \qquad u'(-1) = \beta u(-1),$$

$$\text{even: } \Big[q^2 + \frac{3\eta}{2}\Big(\beta + \frac{3\eta}{2}\Big)\Big]\sin q - \beta q\cos q = 0, \qquad \text{odd: } q\cos q + \Big(\beta + \frac{3\eta}{2}\Big)\sin q = 0, \qquad q^2 = \Lambda - \frac{9\eta^2}{4}.$$

mode ($\eta = 3$)	$\Lambda = (m\,fL)^2$	status
photon, gluons, graviton ($\beta = 0$, flat profile)	0 exactly	DERIVED
first even Kaluza-Klein level	$9\eta^2/4 + \pi^2 = 30.1196$	DERIVED (finite elements 1.9×10^{-10})
first odd level	27.0890 ($q\cot q = -9/2$)	DERIVED (finite elements 4.5×10^{-10})
gap	$9\eta^2/4 = 20.25$	DERIVED

mode ($\eta = 3$)	$\Lambda = (m\,fL)^2$	status
W^+, W^- ($\beta_W = 0.011912$)	1.32149×10^{-5}	DERIVED (fit, finite elements 3×10^{-14})
Z ($\beta_Z = \beta_W/\cos^2\theta_W = 0.015343$, angle of the couplings $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0 = 0.776368$)	1.70150×10^{-5}	DERIVED
Dirichlet bound ($\beta \to \infty$)	0.010016: $m_W \le 0.01824\, m_{KK}$	DERIVED

Here θ_W is the angle of the gauge couplings, $\cos^2\theta_W = \hat{g}^2/(\hat{g}^2 + \hat{g}'^2)$. The bending of the tip-localised profiles moves the mass ratio $m_W^2/m_Z^2 = \Lambda_W/\Lambda_Z$ away from $\cos^2\theta_W$, and the electroweak parameter $\rho = \Lambda_W/(\Lambda_Z \cos^2\theta_W)$ exceeds 1 ($\rho - 1 = 3.2 \times 10^{-8}$ at $\beta_W = 10^{-6}$: custodial symmetry at leading order; 6.4×10^{-5} at $\beta_W = 0.002$, 3.2×10^{-3} at 0.1). This tree-level custodial violation is fitted to the measured $\rho_0 - 1 = 0.00038$ (the violation beyond the Standard-Model loops), and the angle of the couplings is fixed by the measured masses, $\rho = m_W^2/(m_Z^2 \cos^2\theta_W) = \rho_0$, i.e. $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0 = 0.776368$ (on-shell $m_W^2/m_Z^2 = 0.776663$); both masses are then reproduced. With $m_W = 80.3625$ GeV this gives the scales (DERIVED)

quantity	value
Kaluza-Klein mass $m_{KK} = m_W\sqrt{\Lambda_1/\Lambda_W}$	121.32 TeV
$H = m_{KK} f_* \eta/\sqrt{\Lambda_1}$	14.296 TeV
proper half cell fL; period of x_0, $\pi/(6H)$	8.926×10^{-21} m; 7.227×10^{-21} m
gap $3H/(2f_*)$	99.48 TeV
radion mass $3.387H$	48.42 TeV
model $m_Z = m_W\sqrt{\Lambda_Z/\Lambda_W}$	91.1879 GeV (the measured value, reproduced)

5.2 Fermions

$$h_j = j\Big[-\frac{i}{f}\sigma_1\partial_y + \mu\,\sigma_2 + e^{-a_4}k\,\sigma_3\Big],$$

$$D(e) = e^2 s_L s_R - \kappa_f^2 (C_L - c_L s_L)(C_R + c_R s_R) = 0, \qquad s = \frac{\sinh w}{w}, \quad C = \cosh w, \quad w = \sqrt{c^2 - e^2},$$

with k the momentum of the mode along x_1 (not the block variable) and $c = f\mu L$; the first-order solution $e = \kappa_f \zeta(c_L)\zeta(-c_R)$, $\zeta(c)^2 = c\,e^{-c}/\sinh c$ ($c_L > 0$, $c_R < 0$: both zero modes at the fold), agrees with the exact root to 1.2×10^{-8}, and direct shooting of the coupled Dirac system agrees with the determinant to 3.5×10^{-10}. With $\kappa_f = 1$ and $c_L = -c_R = c$ (DERIVED):

fermion	mass	c	μ/H	proper width $fL/(2c)$	$e = m\,fL$
ν	0.05 eV	18.6714	28.873	2.39×10^{-22} m	2.262×10^{-15}
e	0.511 MeV	10.3042	15.934	4.33×10^{-22} m	2.312×10^{-8}
u	2.16 MeV	9.5452	14.761	4.68×10^{-22} m	9.771×10^{-8}
d	4.70 MeV	9.1345	14.126	4.89×10^{-22} m	2.126×10^{-7}

5.3 dirac16complex

With $M = m$, fold condition $\chi_2(0) = 0$ (even) or $\chi_1(0) = 0$ (odd), tip $\chi_2(-L) = 0$ and $c_\psi = f_* mL = 0.646667$, the exact $k = 0$ spectrum is: the fold zero mode ($\varepsilon = 0$, all eight blocks); even bulk levels $\sqrt{m^2 + (n\pi/(fL))^2}$, the lowest $4.959981\,m$; odd levels $\sqrt{m^2 + p^2}$ with $\tan(p\,fL) = -p/m$, the lowest $3.102204\,m$ ($p = 2.93661\,m$, $p\,fL = 1.89901$); and the fold band $\varepsilon = e^{-a_4}|k|$ (massless, $\varepsilon/|k| = 1$ to 6×10^{-14}) (DERIVED).

6 Step 3: Kohn-Sham states of dirac16complex coupled to all fields

6.1 The Kohn-Sham equations

$$h_j\chi_o=\varepsilon_o\chi_o,\qquad M_{eff}=m+\frac{15}{16}\lambda S+\frac{\langle\mathcal{F}\rangle}{\Lambda_\star^3},\qquad v=-\frac{\lambda}{16}n+\frac{c_V}{\Lambda_\star^6}\sum_\psi q_\psi n_\psi+v_{I9},$$

$$n=\sum wgf_on_o,\quad S=\sum wgf_os_o,\quad n_o=P\chi_o^\dagger\chi_o,\quad s_o=P\chi_o^\dagger j\sigma_2\chi_o,\quad t_o=P\chi_o^\dagger j\sigma_3\chi_o,\quad P=\frac{e^{-3Hy}}{\mathrm{Vol}_7},$$

$$E_{KS}=\sum gf_o\varepsilon_o-2\mathrm{Vol}_7\int e^{3Hy}e_{int}dy,\qquad e_{int}=\frac{15}{32}\lambda S^2-\frac{1}{32}\lambda n^2,$$

$$\rho=\sum wgf_o\varepsilon_on_o-e_{int},\qquad p_3=\sum wgf_o\tfrac{e^{-a_4|k|}}{3}t_o+e_{int},\qquad p_t=e_{int},$$

$$p_0=\sum wgf_o[(\varepsilon_o-v)n_o-M_{eff}s_o-e^{-a_4}|k|t_o]+e_{int},$$

with $w=1/2$ per fold parity, $g=4r_3(|\vec{n}|^2)$ for the momentum shell $\vec{k}=\Delta k\,\vec{n}$ ($\vec{n}\in\mathbb{Z}^3$, $r_3(|\vec{n}|^2)$ the number of representations of $|\vec{n}|^2$ as a sum of three squares, $\Delta k=0.25$; here k is the momentum, not the block variable), occupations f_o, and $\mathrm{Vol}_7=(2\pi/\Delta k)^3v_tf_*^4=34.27$ ($v_t=1$, $a_4=0$; the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1-e^{-3\eta})/(3H)$). The orbitals are expanded in the exact free eigenfunctions of each sector (60, 120, 240 per sector), iterated to self-consistency (tolerance 10^{-13}, safeguarded Anderson mixing), and extrapolated in the basis size (Richardson).

6.2 Validation

The independent solver reproduces the record's two reference states (at $f=1$, $a_4=0$): $N=8$, $\lambda=157.7$: $E_{KS}=-0.037786682923$ (reference -0.037786682923), HOMO and LUMO to 10^{-14}, integrated $p_0=-0.08860537545$ (reference -0.08860537548); $N=592$, $\lambda=2.131$: $E_{KS}=358.30104644787$ (identical), integrated $p_0=-6.7691415097$ (reference -6.7691415108) (DERIVED).

6.3 Ground states on the stabilised block

Two families of states (the eight $k=0$ fold zero modes, $N=8$; a fold-band gas, $N=592$) with $\lambda=0$, the benchmark $\pm\lambda_B$ ($\lambda_B=7\times10^{-5}H^{-6}$: mean fields below 5 % of m) and the record's coupling scaled to the block (stress test, $\lambda=157.7f_*^4=0.3405$ and $2.131f_*^4=0.00460$) (DERIVED; 120 functions per sector, Richardson errors below 1.5×10^{-9}):

state	E_{KS}	E_{int}	HOMO	LUMO	iterations
$N=8$, $\lambda=0$	0	0	0	0.25	1
$N=8$, $+\lambda_B$	-2.053×10^{-4}	-2.053×10^{-4}	-5.13×10^{-5}	0.24995	14
$N=8$, $-\lambda_B$	2.053×10^{-4}	2.053×10^{-4}	5.13×10^{-5}	0.25005	14
$N=8$, stress	-0.70123	-0.42170	-0.14037	0.10859	61
$N=592$, $\lambda=0$	361.11614	0	0.79057	0.82916	1
$N=592$, $+\lambda_B$	359.99060	-1.12646	0.78676	0.82535	14
$N=592$, $-\lambda_B$	362.23906	1.12121	0.79436	0.83295	13
$N=592$, stress	307.67139	-32.2217	0.64506	0.68347	56

state	$\int p_3$	$\int p_t$	$\int p_0$	max $\lvert\delta M\rvert$	max $\lvert v\rvert$
$N=8$, $+\lambda_B$	-2.05×10^{-4}	-2.05×10^{-4}	-4.447×10^{-4}	2.4×10^{-7}	6.7×10^{-4}
$N=8$, $-\lambda_B$	2.05×10^{-4}	2.05×10^{-4}	4.447×10^{-4}	2.4×10^{-7}	6.7×10^{-4}
$N=8$, stress	-0.4217	-0.4217	-1.30076	1.98	1.83
$N=592$, $\lambda=0$	120.372	0	8.6×10^{-12}	0	0

state	$\int p_3$	$\int p_t$	$\int p_0$	max $\lvert\delta M\rvert$	max $\lvert v\rvert$
$N = 592$, $+\lambda_B$	119.245	-1.1265	-2.43688	1.3×10^{-3}	0.050
$N = 592$, $-\lambda_B$	121.493	1.1212	2.43174	1.3×10^{-3}	0.050
$N = 592$, stress	87.614	-32.222	-97.8188	2.09	1.88

(The integrals are $2\mathrm{Vol}_7 \int e^{3Hy}(\cdot)\, dy$; the $N = 8$, $\lambda = 0$ state has all of them zero.)

Checks (DERIVED): $E_{KS} = 2\mathrm{Vol}_7 \int e^{3Hy}\rho\, dy$ to 4×10^{-15}; the y-conservation law $[e^{3Hy}p_0]^0_{-L} = 3H \int e^{3Hy}p_t dy$ to 9×10^{-8} (Richardson; the raw residual falls like n_b^{-3}); observed basis orders 4.95 (energy) and 2.97 (hidden pressure); the free gas has $w = \int p_3 / \int \rho = 1/3$ exactly (four-dimensional radiation). The self-interaction shifts the fold-band levels almost rigidly (the spacing between $k = 0$ and the first shell is 0.25 free, 0.25000 at λ_B, 0.2490 in the $N = 8$ stress state). A scalar mean field cannot gap the fold band at $k = 0$: $\chi = (e^f \int^{M_{eff}dy}, 0)$ solves $j[-\frac{i}{f}\sigma_1\partial_y + M_{eff}\sigma_2]\chi = 0$ with both boundary conditions for every $M_{eff}(y)$.

6.4 Dark-matter quanta

Eight quanta in the lowest odd level at $k = 0$ on the $N = 8$ ground state add exactly $8\varepsilon_{DM} = 24.817633063$ (computed 24.817633063) at $\lambda = 0$ and 24.81833 at $+\lambda_B$; they are cold, $w = p_3/\rho = 0$ (2.8×10^{-5} with λ_B); their hidden pressure is $p_0/\rho = 0.772034343$, equal to $-d\ln\varepsilon_{DM}/d\ln f = 0.772034343$ obtained independently from the x_0 eigenvalue (DERIVED): the dark-matter mass is an x_0 eigenvalue, and it pushes on the block.

6.5 The Standard-Model fields acting on dirac16complex

The portal mean fields were applied with amplitude e and shapes given by the sources (the electron and up-quark zero-mode densities times P for I11; a uniform gluon condensate for I12; a tip-localised Z charge on the upper component for I9). Linear responses of the lowest odd level (DERIVED; equal to the Hellmann-Feynman expectation $d\varepsilon/de = \langle\chi|\partial h/\partial e|\chi\rangle$ to 10^{-8}):

portal mean field	$d\varepsilon_{odd}/de$	change of the fold-band gap
I11, electron density (v)	0.507961	1.6×10^{-10}
I11, up-quark density (v)	0.548528	2.1×10^{-10}
I12, gluon condensate (uniform M)	0.707196	2.9×10^{-11}
I9, tip Z charge (upper component): Ψ_A odd level; Ψ_B (the dark matter)	0.259489; exactly 0	7.9×10^{-11}

The fold band stays massless under every portal. Every orbital has $\chi_2(-L) = 0$, so the scalar density $\bar\Psi\Psi = 2s_2ab$ vanishes at the tip: no tip operator built on $\bar\Psi\Psi$ reaches dirac16complex, and I9 acts on the upper component only. For Ψ_B ($B = +1$) the tip projector $P_+ = (1 - B\gamma^{x_0})/2$ annihilates every orbital at the tip, so the Ψ_B dark matter has no I9 coupling at all (section 7). The I12 response defines the four-dimensional coefficient of the dark-matter operator: $G_i = \mathcal{O}\, c_i g_i^2/\Lambda_\star^3$ with $\mathcal{O} = 0.7072$.

6.6 Back-reaction on the block

The block-averaged hidden pressure $\bar p = (p_0 + 3p_t)/4$ of a Kohn-Sham state displaces the block by $\delta k = \kappa\bar p/(3H^2 - 4\kappa\rho_b^*/k_*) = -0.140028\,\kappa\bar p/H^2$ (stiffness $-7.1414H^2$), without changing the four-dimensional cosmological term at linear order ($\frac{d}{dk}[6H^2k(1 - 2w_2k) + \kappa\rho_b] = -6 \times 10^{-12}$ at k_*). Cell averages (units of the Kohn-Sham calculation, $H = 1$): $\bar p = -1.16 \times 10^{-5}$ ($N = 8$, $+\lambda_B$), -0.0636 ($N = 592$, $+\lambda_B$), ≈ 0 (free gas); the shift is negligible for any realistic density (relative size $(A/H)^2$). The difference $p_0 - p_t$ (-0.0574 for $N = 592$, $+\lambda_B$) drives a block-shape mode outside the order-0 metric (OPEN).

7 Step 4: the electroweak sector, N_{eff} and the CNB

Charges and couplings (DERIVED): $Q = T_3 + Y$ gives $(u,d)_L = (2/3, -1/3)$, $(\nu, e)_L = (0, -1)$, $u_R = 2/3$, $d_R = -1/3$, $e_R = -1$, $\nu_R = 0$; $[T_3, T_\pm] = \pm T_\pm$ with $Y(W) = 0$: $W^+ = (W^1 - iW^2)/\sqrt{2}$ **carries** $+e$, $W^- = (W^1 + iW^2)/\sqrt{2}$ **carries** $-e$, and W^- is the antiparticle of W^+; the photon couples with $e = gg'/\sqrt{g^2 + g'^2}$ times Q, the Z with $\sqrt{g^2 + g'^2}(T_3 - Q\sin^2\theta_W)$; the one generation is anomaly free ($SU(3)^2U(1)$, $SU(2)^2U(1)$, $U(1)^3$, gravitational, $SU(3)^3$ all vanish; four doublets, no Witten anomaly).

The two halves of dirac16complex. B commutes with the four-dimensional Lorentz generators and with C, so $\Psi = \Psi_A + \Psi_B$ ($B = -1$, $B = +1$) splits into Lorentz-invariant halves and $\bar\Psi\Psi = \bar\Psi_A\Psi_A + \bar\Psi_B\Psi_B$; the identity $\Gamma_4 = -B\gamma^{x_0}$ (DERIVED, 16×16) gives $P_+ = (1 - B\gamma^{x_0})/2$, which equals $(1 + \gamma^{x_0})/2$ on Ψ_A and $(1 - \gamma^{x_0})/2$ on Ψ_B; the latter annihilates every orbital at the tip. Hence the tip Z charge densities (DERIVED): Ψ_A fold band $a_0(-L)^2 = 0.163$ per unit length in each of its four blocks, Ψ_B fold band 0; Ψ_A odd (heavy) states 0.257, Ψ_B odd (dark-matter) states 0, and every transition current $\bar\Psi_A\gamma^\mu P_+\Psi_B$ vanishes at the tip.

The Z invisible width. With one neutrino and the two Ψ_A Weyl species of Z-coupling r times the neutrino's,

$$N_\nu = \frac{\Gamma_{inv}}{\Gamma(Z \to \nu\bar\nu)} = 1 + 2r^2 = 2.9840 \pm 0.0082 \implies r^2 = 0.9920 \pm 0.0041, \qquad r = 0.99599.$$

N_{eff} and the CNB. The neutral-current rate scales as $r^2 G_F^2 T^5$ against $H \propto T^2/M_{Pl}$, so Ψ_A decouples at $T \approx 2.6\,\mathrm{MeV} \times r^{-2/3} = 2.61$ MeV, after the QCD and pion entropy releases; $T_{\Psi_A} = T_\nu$ for every $r^2 > 2.2 \times 10^{-3}$, and

$$N_{eff} = 3.044, \qquad T_\nu = \Big(\frac{4}{11}\Big)^{1/3} T_\gamma = 1.9454\ \mathrm{K}, \qquad n = \frac{3}{4}\frac{\zeta(3)}{\pi^2}\,2T_\nu^3 = 112.0\ \mathrm{cm}^{-3}\ \text{per Weyl species.}$$

The cosmic neutrino background of UM-2: ν_e (0.05 eV, non-relativistic today) and two massless Ψ_A species, 336 cm^{-3} in all (DERIVED). The Ψ_B fold band has no tip coupling and no scalar density and, with $c_V = 0$, is never thermalised.

The Z width of one generation (tree level, $\Gamma(Z \to f\bar f) = N_c G_F m_Z^3 (g_V^2 + g_A^2)/(6\sqrt{2}\pi)$, $g_V = T_3 - 2Q\sin^2\theta_W$ with the $\overline{MS}$ value $\sin^2\theta_W = 0.23122$, $g_A = T_3$):

channel	ν_e	e	u	d	Ψ_A (2 Weyl)	total	measured
width (MeV)	165.9	83.4	285.4	367.9	329.1	1231.7	2495.5

The $\overline{MS}$ angle absorbs the Standard-Model loop corrections; with the tree-level angle of the couplings of the mass fit, $\sin^2\theta_W = 0.22363$, the widths of e, u, d would be larger by 0.55 %, 1.39 % and 0.95 % and the total would be 1239.6 MeV (DERIVED): a per-cent scheme ambiguity, far below the factor-two conflict of the total width.

The invisible width is reproduced through the fit of N_ν (tree level 495 MeV against the measured 499.0 ± 1.5 MeV); the total is not, because the order restricts the particle content to one generation (the μ, τ, s, c, b channels are missing): an inherent conflict of the specified content with data.

8 Step 5: the CMB, the CNB and the BAO

The four-dimensional effective theory is general relativity (step 1) with photons, baryons, ν_e, the two Ψ_A species, the Ψ_B dark matter and Λ_4. Its linear perturbations obey the standard Einstein-Boltzmann equations (Newtonian gauge, conformal time, $ds^2 = a^2[-(1+2\psi)d\tau^2 + (1-2\phi)d\vec{x}^{\,2}]$):

$$k^2\phi + 3\mathcal{H}(\phi' + \mathcal{H}\psi) = -4\pi G_N a^2\delta\rho, \qquad k^2(\phi' + \mathcal{H}\psi) = 4\pi G_N a^2(\bar\rho + \bar P)\theta,$$

$$\phi'' + \mathcal{H}(\psi' + 2\phi') + (2\mathcal{H}' + \mathcal{H}^2)\psi + \frac{k^2}{3}(\phi - \psi) = 4\pi G_N a^2\delta P, \qquad k^2(\phi - \psi) = 12\pi G_N a^2(\bar\rho + \bar P)\sigma,$$

with the cold dark matter $\delta_c' = -\theta_c + 3\phi'$, $\theta_c' = -\mathcal{H}\theta_c + k^2\psi$, the baryon-photon fluid with Thomson scattering, the massless hierarchy for Ψ_A and the massive hierarchy for ν_e. They were solved with CAMB 2.0.5 at the Planck 2018 values $\omega_b = 0.02237$, $\omega_c = 0.1200$, $100\theta_{MC} = 1.04092$, $\tau_{reio} = 0.0544$ (the optical depth to reionisation), $\ln(10^{10}A_s) = 3.044$, $n_s = 0.9649$ (θ held fixed, so H_0 is an output; H_0 in km/s/Mpc). The models are Planck 2018 ΛCDM (three neutrinos, $\sum m_\nu = 0.06$ eV), UM-2 (ν_e of 0.05 eV and two massless Ψ_A species) and the UM-2 variant with the Ψ_B fold band thermalised above the QCD crossover (DERIVED):

model	H_0	Ω_m	N_{eff}	Y_{He}	age (Gyr)	z_*	r_{drag} (Mpc)	BAO χ^2 (13 points)
ΛCDM	67.34	0.3154	3.044	0.24566	13.799	1089.91	147.103	32.55
UM-2	67.43	0.3143	3.044	0.24566	13.793	1089.91	147.105	30.31
UM-2, Ψ_B hot	70.62	0.2865	3.318	0.24925	13.515	1090.17	145.750	15.97

The UM-2 TT and EE spectra agree with Planck ΛCDM to 2.2×10^{-4} and 3.7×10^{-4} ($30 \le \ell \le 2500$), for TT at most 0.011 of the cosmic variance $\sqrt{2/(2\ell+1)}$ of a single multipole: UM-2 reproduces the CMB of ΛCDM. If the Ψ_B fold band were thermalised and decoupled just above the QCD crossover, it would add $\Delta N_{eff} = 2(10.75/g_s)^{4/3} = 0.274$, with $g_s = 47.75$ counting the plasma it leaves (photons, gluons, u, d, e, ν_e and the two Ψ_A species) and 10.75 the same count at neutrino decoupling (0.218 for decoupling above the electroweak scale, $g_s = 56.75$). At fixed Planck parameters this variant changes TT by up to 4.2 % for $30 \le \ell \le 2500$: the largest change, −4.2 % at the upper end $\ell = 2500$ of the range (the change still grows into the damping tail), is 2.1 cosmic-variance units of a single multipole; at $\ell = 2$ the change is 4.9 %, where the cosmic variance of one multipole is 63 %; the full-sky cosmic-variance $\chi^2 = \sum_b (\bar\delta_b/\sigma_b)^2$ over bins b of $\Delta\ell = 50$, with $\bar\delta_b$ the mean relative deviation and $\sigma_b = \sqrt{2/\sum_{\ell\in b}(2\ell+1)}$, is 2314; it is not a fit to the CMB, and a joint CMB and BAO fit of it is OPEN (it is not excluded by this computation). The CNB enters through $N_{eff} = 3.044$ and the ν_e mass.

9 Step 6: the dark-matter relic and its detection

The lightest odd state of Ψ_B is a Dirac fermion of mass $M_{DM} = 3.102204\,m = 44.35$ TeV ($m = H$), cold, neutral, with no tree-level Z coupling. Through I12 it annihilates in the p-wave into gauge-boson pairs: for $\mathcal{L} = G\,\bar\chi\chi F_{\mu\nu}F^{\mu\nu}$ per abelian channel, $\sigma v = G^2 s(s-4M^2)/(4\pi) \to G^2M^4v^2/\pi$, in total $\sigma v = (M^4v^2/\pi)(8G_g^2 + 3G_W^2 + G_B^2)$ (the polarisation sum $\sum|(k_1\cdot k_2)(e_1\cdot e_2) - (k_1\cdot e_2)(k_2\cdot e_1)|^2 = s^2/2$ and the spin trace $\mathrm{Tr}[(\not p_1 + M)(\not p_2 - M)] = 2(s-4M^2)$ checked numerically). Freeze-out (Dirac fermion, $g_* = 56.75$: the gauge bosons without a Higgs, e, ν_L, u, d and the two Ψ_A species; ν_R never thermalised):

$$x_f = \ln\left[0.038\,c(c+2)\,g\,M_{Pl}M\langle\sigma v\rangle/\sqrt{g_* x_f}\right] = 30.31, \qquad \Omega h^2 = \frac{2\times1.07\times10^9\,\mathrm{GeV}^{-1}\,x_f}{\sqrt{g_*}\,M_{Pl}\,3b/x_f} = 0.120$$

($\langle\sigma v\rangle = 6b/x$, $c = 1/2$, $g = 2$, $M_{Pl} = G_N^{-1/2} = 1.22\times10^{19}$ GeV, standard radiation-dominated freeze-out) requires $\langle\sigma v\rangle(x_f) = 1.37\times10^{-25}$ cm^3/s, below the s-wave thermal unitarity estimate $(4\pi/M^2)\sqrt{x_f/\pi} = 2.3\times10^{-25}$ cm^3/s (an estimate made for an s-wave process, applied here to a p-wave one), i.e.

$$(8G_g^2 + 3G_W^2 + G_B^2)^{1/2} = (16.6\ \mathrm{TeV})^{-3}, \qquad G_i = \mathcal{O}\,c_i g_i^2/\Lambda_\star^3, \quad \mathcal{O} = 0.7072.$$

Direct detection (ESTIMATE): even if the whole annihilation went into gluons, the trace anomaly ($\langle N|G^\alpha G^\alpha|N\rangle = -\frac{8\pi}{9\alpha_s} m_N f_{TG}$) gives $\sigma_{SI} = \mu^2 f_N^2/\pi = 4.3\times10^{-53}$ cm^2, eight orders below the LZ limit extrapolated to 44 TeV (2.4×10^{-45} cm^2); the electroweak channels have no tree-level nucleon coupling. The relic value is conditional: if the radion oscillation left over from the deflation (which redshifts like dust) dominated the Universe until it decayed, it would dilute the dark matter after freeze-out (the moduli problem, below).

10 Accuracy of the solution

- Exact (symbolic): the background equations, the static block, $G(k_*)$, the radion frequency, the first-order block response, the dimensional reduction, the field reductions and the zero modes.

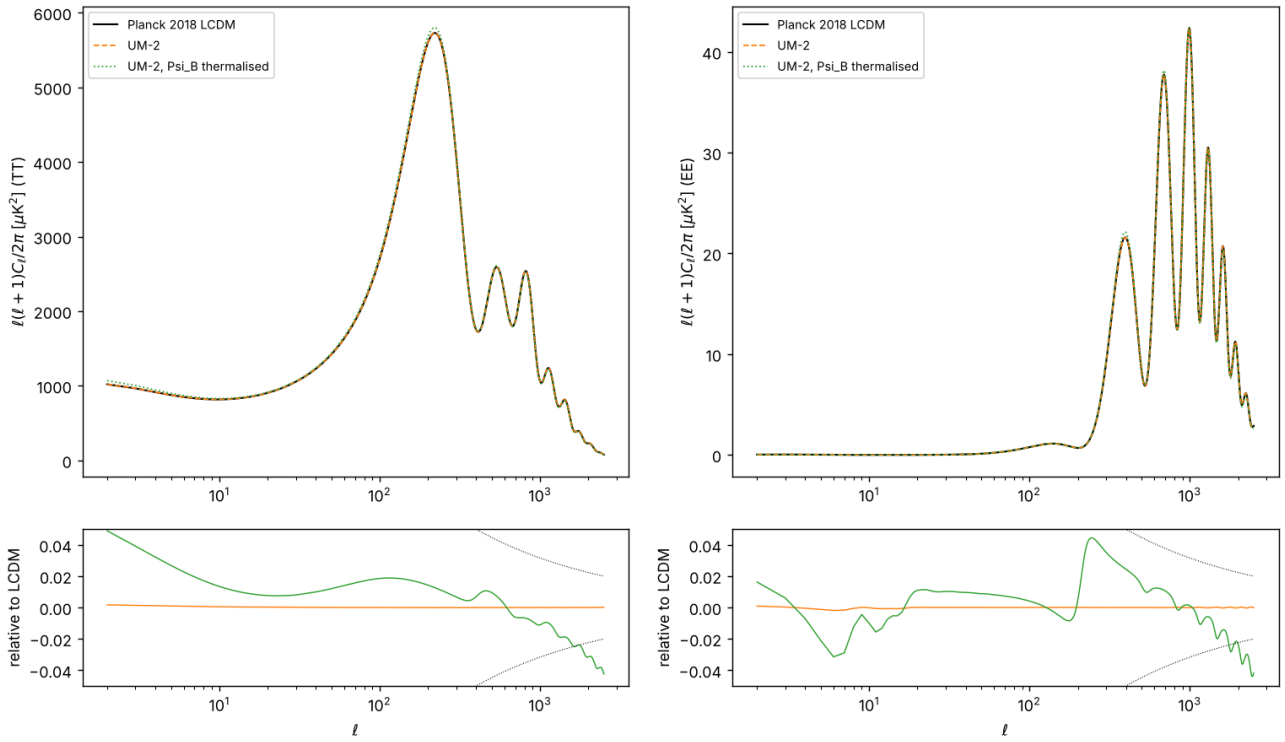


Figure 2: CMB temperature (left) and E-mode (right) spectra of UM-2, of Planck 2018 ΛCDM and of the variant with a thermalised Ψ_B fold band; lower panels: relative difference to ΛCDM with the cosmic variance of one multipole (dotted).

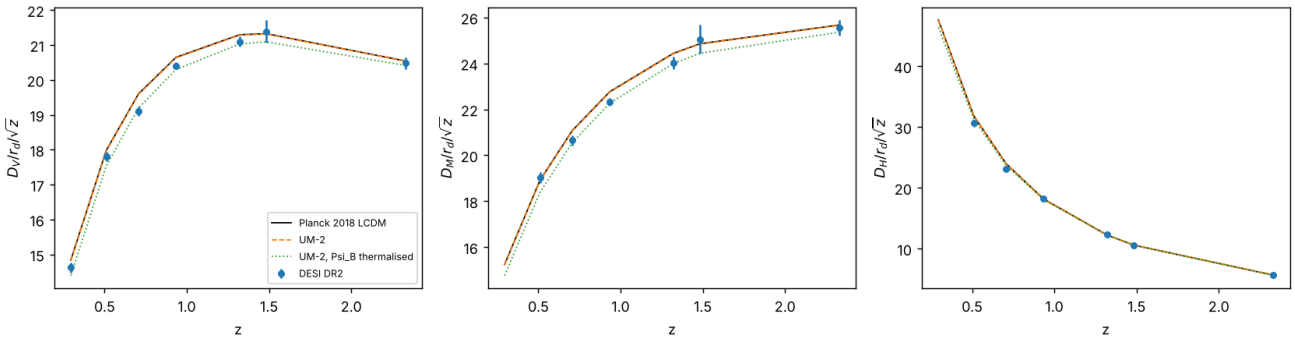


Figure 3: BAO distances divided by $\sqrt{z}$ against the DESI DR2 measurements; the χ^2 uses D_V for BGS and D_M, D_H with their correlation for the other tracers (the D_V points of the anisotropic tracers are shown for illustration).

- Numerical, with measured errors: boson levels (mpmath, 30 digits; finite elements $\leq 5 \times 10^{-10}$); fermion masses (exact roots; first order 1.2×10^{-8}; shooting 3.5×10^{-10}); Kohn-Sham energies (Richardson errors below 1.5×10^{-9} relative; conservation 9×10^{-8}); primordial integrations (constraint below 6×10^{-10}); CMB (CAMB default accuracy, lensing included).
- Approximations: the static block with an assumed stabiliser; the good sector (no extra-time momenta); the Kohn-Sham mean-field treatment (local exchange of the contact interaction); portal couplings as external mean fields; tree level for the electroweak widths; the freeze-out formula; order ϵ for the cosmology. A fresh-clone rerun (2026-10-18) passed all 182 checks and reproduced every stored number to 10^{-9} relative or better.

11 What the solution does not reproduce, and what is open

- **Conflicts with data, inherent in the order**: no Higgs boson (one is observed at 125 GeV), and without it the longitudinal W^+W^- scattering violates perturbative unitarity near $\sqrt{8\pi}\, v \approx 1.2$ TeV, far below the Kaluza-Klein modes; the total Z width of one generation, 1.23 GeV against 2.4955 GeV.
- **Open**: the origin of the stabiliser; the extra-time modes outside the good sector (tachyonic for momentum $q > fM$); the closed time-like curves implied by compact extra times; the radion left over from the deflation (ESTIMATE: a Planck-suppressed decay $\Gamma = C_r\, m_r^3/(4\pi \bar{M}_{Pl}^2)$, with the reduced Planck mass $\bar{M}_{Pl} = 2.435 \times 10^{18}$ GeV and an order-one coupling factor C_r, gives the lifetime $0.43\,\mathrm{s}/C_r$ and the reheating temperature $T_{RH} = (90/(\pi^2 g_*))^{1/4}\sqrt{\Gamma \bar{M}_{Pl}} = 1.85\sqrt{C_r}$ MeV, $g_* = 10.75$, below the 4.7 MeV that nucleosynthesis needs unless $C_r > 6.5$: a moduli problem); the block-shape mode driven by $p_0 - p_t$; the tuning of Λ_4 ($10^{-92} H^2$); the Kaluza-Klein graviton spectrum with the Lovelock corrections; the electroweak S parameter of the Higgsless tip; anomalies of the eight-dimensional bulk theory; the joint fit of the thermalised-Ψ_B variant.